

Hyper parameter Tuning using Optuna:

Random Forest:

max-depth & n-estimators.

GridSearch.

- Cal all the parameter.

	n-estim		
	50	100	150
max-depth	1		
	2		
	3		

Random Search

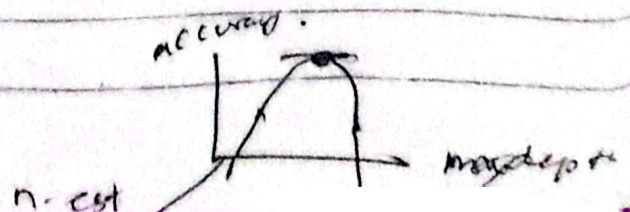
choose random combination to try.

Optuna: [Bayesian Search]

there is relation betⁿ accuracy, max-depth and n-estimator.

- Try some combinations. ~ 10
- accuracy = $f(\text{max-depth}, \text{n-estimator})$ - obj. fⁿ
- Find out the nature and, so ~~to~~ it will search for maxima [for max value of accuracy]

[By using Bayesian Optimization]



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Optuna: direction = 'maximize'
or 'minimize'

Sampler = which value to choose next

Study: best trial value

Study: best-trial params

- We can do random search & grid search in the Optuna.
- We can also see the visualization also from optuna.visualization.

It also gives which ~~hyperparameter~~ parameter is more imp. in percentage in graph.

Define by run [Dynamic Search Space].

50:00

✓ Optuna gives which hyperparameter is best, moreover, it also gives which algorithm is best.

• model as parameter.

• fast → distributed computations.